RICE UNIVERSITY OUTCOMES ASSESSMENT REPORTING (ROAR) HANDBOOK

Section I: Introduction to the Handbook

The purpose of this handbook is to provide you with an orientation to and an overview of academic program outcome assessment reporting as it is conducted at Rice University. The assessment process at Rice is referred to as Rice Outcomes Assessment Reporting (ROAR). This handbook aims to provide information, resources, and examples to assist you in preparing for assessment activities, creating an assessment plan, and documenting your assessment and continuous improvement related activities.

Layout of the Handbook

The handbook is divided into multiple sections.

- Section I (page 1) describes the purpose of the handbook and sections of the handbook.
- Section II (pages 1-4) describes the history of outcomes assessment at Rice University and explains how outcomes assessment fits into the larger scope of institutional effectiveness. You will find useful definitions as well as a "big picture" overview of ROAR in this section.
- Section III (pages 4-15) offers guidance in preparing for assessment, including a description of what procedures should be in place to formulate the annual ROAR and Follow Up Reports, and a description of the annual reporting process.
- Section IV (page 15) provides direction on where you can find help to support assessment work.
- The Appendix includes a list of Bloom's Taxonomy Action Verbs (page 16), a Glossary of Terms (pages 17-24) and list of Works Cited (page 24-27).

Section II: Assessment Overview

The Southern Association of Colleges and Schools Commission on Colleges (SACSCOC), Rice University's institutional accreditor, explains that: "Effective institutions demonstrate a commitment to the principles of continuous improvement. These principles are based on a systematic and documented process of assessing institutional performance with respect to mission in all aspects of the institution. An institutional planning and effectiveness process involves all programs, services and constituencies [and] is linked to the decision-making process at all levels" (SACSCOC, 2023, p. 19).

Institutional effectiveness at Rice is part of an integrated model that considers course/activity and program/unit level assessment (ROAR) alongside other data, to comprehensively assess

achievement at all levels of the institution. These various elements interact and interrelate in numerous ways, to support the broader institutional strategic priorities and reflect the data-informed culture at Rice.

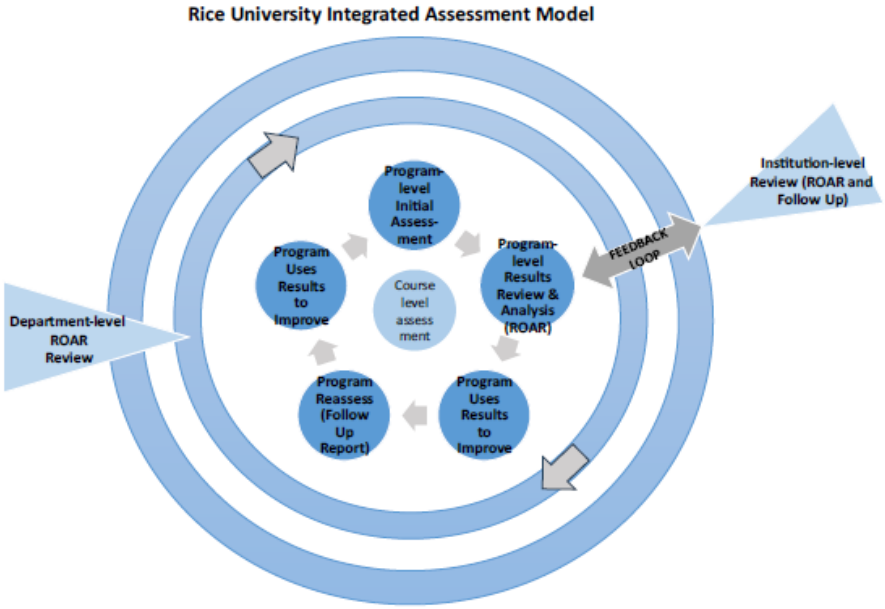

Figure 1. Rice Integrated Institution-wide Assessment Model

A Brief History of Assessment at Rice University

The Office of Institutional Effectiveness (OIE) was created in March 2007 with one of several charges being oversight of the centralized outcomes assessment process. Since then, academic departments have followed a cycle of activities consistent with the philosophy of institutional effectiveness (i.e., identifying outcomes, assessing student achievement of these outcomes, developing improvement plans based on assessment results, and evaluating those plans). Administrative units also participate in the annual outcomes assessment process by setting goals at the outset of a fiscal year and submitting reports at the end of the year. OIE collaborated with appropriate campus partners including but not limited to the Faculty Committee on Undergraduate Curriculum, the Graduate Council and the Faculty Senate to ensure all new degree programs include student-focused program learning outcomes (PLOs) and assessment plans. These requirements are outlined in the document entitled: Requirements for the Creation, Modification, Oversight, and Elimination of Credentialed Undergraduate Academic Programs at Rice University (see pg. 13).

The Rice outcomes assessment reporting process began in spring 2006. At that time, academic units developed student learning outcomes and preliminary assessment plans, and actual

assessments began during the 2006-2007 academic year. The first complete reports were submitted to OIE in 2008. At that time, academic programs summarized their assessment activities using a two-part form: part one was called the Rice Outcomes Assessment Report (ROAR), and part two was called the Report on Improvement Plan's Effectiveness (RIPE). Then, as it is today, the PLOs were published in the General Announcements (GA; Rice's University Catalog) and are reviewed annually by programs and updated regularly. PLO updates were reviewed by the Office of the Registrar and OIE prior to publication in the GA. OIE worked with programs to update Assessment Plans to reflect edits to published PLOs and corresponding assessment methodology.

In recent years, the RIPE has evolved to simply being referred to as a Follow-up Report on which programs and units document improvement actions they have implemented based on assessment results. In 2026, OIE became the Institutional Effectiveness team, a part of the broader IDEAS office at Rice (Institutional Data, Evaluation, Analytics and Strategy) which also includes the Institutional Research and Analytics team. Then, as today, academic programs and administrative units submit ROARs and Follow-up Reports annually to IE—academic ROARs and Follow-up Reports are submitted each June, and administrative units submit their reports in August. Historically, academic programs have completed a full cycle of assessment for all PLOs (to include the initial ROAR and at least one Follow-up Report) within a three-year period.

As of spring 2026, IE has updated the ROAR reporting process for both academic programs and administrative units. Academic programs and administrative units follow a schedule of reporting for even numbered PLOs/Goals during even numbered years, and odd numbered PLOs/Goals during odd numbered years. At least one Follow-Up Report will be submitted each year which describes a program improvement made during the academic year, based on systematically gathered and analyzed assessment data. For example, if a program/unit has 4 PLOs/Goals, in spring 2027 initial ROARs for PLO/Goals 1 and 3 would be submitted, along with a Follow-Up Report. Additionally, in Spring 2026 IE introduced the use of the online platform, Nuventive Improve, for assessment purposes. Assessment personnel submit Initial ROAR reports and Follow-Up reports into this platform which serves as a centralized assessment data repository for all academic programs and administrative units at Rice.

As of spring 2026, IE's assessment team consists of three full time staff members who are trained in outcomes assessment and who assist programs and units in developing outcomes, determining appropriate methods, creating, and updating assessment plans, and provide 1:1 and small and large group training on how to complete ROARs and Follow-up Reports.

SACSCOC Expectations for Outcomes Assessment

SACSCOC has strongly emphasized continuous improvement and institutional effectiveness in the compliance process. SACSCOC's most recent edition of The Principles of Accreditation: Foundations for Quality Enhancement (January 2024) includes the following Sections and Standards which focus specifically on continuous improvement and institutional effectiveness (below, CR indicates that a standard is a Core Requirement, meaning, it holds especially high importance in the accreditation review process and findings of non-compliance have higher impact):

- *SECTION 7: Institutional Planning and Effectiveness*

Effective institutions demonstrate a commitment to the principles of continuous improvement. These principles are based on a systematic and documented process of assessing institutional performance with respect to mission in all aspects of the institution. An institutional planning and effectiveness process involves all programs, services, and constituencies; is linked to the decision-making process at all levels; and provides a sound basis for budgetary decisions and resource allocations. The Quality Enhancement Plan (QEP) is an integral component of the reaffirmation of accreditation process and is derived from an institution's ongoing comprehensive planning and evaluation processes. It reflects and affirms a commitment to enhance overall institutional quality and effectiveness by focusing on an issue the institution considers important to improving student learning outcomes and/or student success.

- *Standard 7.1: The institution engages in ongoing, comprehensive, and integrated research-based planning and evaluation processes that (a) focus on institutional quality and effectiveness and (b) incorporate a systematic review of institutional goals and outcomes consistent with its mission. (Institutional Planning) [CR]*

- *Standard 7.3: The institution identifies expected outcomes of its administrative support services and demonstrates the extent to which the outcomes are achieved. (Administrative effectiveness).*

- *SECTION 8: Student Achievement*

Student learning and student success are at the core of the mission of all institutions of higher learning. Effective institutions focus on the design and improvement of educational experiences to enhance student learning and support student learning outcomes for its educational programs. To meet the goals of educational programs, an institution provides appropriate academic and student services to support student success.

- *Standard 8.2: The institution identifies expected outcomes, assesses the extent to which it achieves these outcomes, and provides evidence of seeking improvement based on analysis of the results in the areas below: a. student learning outcomes for each of its educational programs. (Student outcomes: educational programs) b. student learning outcomes for collegiate-level general education competencies of its undergraduate degree programs. (Student outcomes: general education) c. academic and student services that support student success. (Student outcomes: academic and student services)*

The compliance elements of the standards presented above are reflected in Rice University's annual ROAR process. More in depth discussions of each standard are provided in the SACSCOC Principles of Accreditation Resource Manual.

Section III: Preparing for Assessment and the Assessment Process

Although outcomes assessment has a substantial history at Rice, confusion and uncertainty about the purpose and function of outcomes assessment is not uncommon, especially among new

faculty and staff members. This section is intended to provide a common base of understanding about assessment, both in general and as it is practiced at Rice.

Assessment Defined

Outcomes assessment is a systematic and ongoing process of gathering and interpreting information to discover if programs/services are meeting intended outcomes and then using the information to enhance/improve the programs/services (adapted from Virginia Commonwealth, 2002, and Marchese, 1987).

Assessment of student learning outcomes (SLOs) and program learning outcomes (PLOs) is a learner-centered process ensuring that students are learning what we intend for them to learn. Palomba and Banta (1999, p. 4) indicate that assessment helps us answer the following key questions about student learning:

- What should graduates know, be able to do, and value?
- Have the graduates of our institution acquired this learning?
- What are the contributions of the institution and its programs to student growth?
- How can student learning be improved?

At Rice, PLOs are defined for each program and are published in the GA. The PLOs are reviewed annually and updated periodically during an annual process managed by the Office of the Registrar. IE also reviews edits to PLOs. These PLOs are the starting point for a program to create a new assessment plan (the strategy by which a program will evaluate student attainment of knowledge, skills and/or abilities—KSAs--using direct methods). The assessment plan is reviewed and updated each time PLOs are updated. Additionally, programs can adjust their assessment plans (methodologies, action steps, etc.) as they see fit, and communicate these changes to IE and within their ROAR reports. Assessment occurs annually and is iterative, with results being used to affect program improvement, and are subsequently re-assessed to determine impact. Rice ROAR reporters use the online platform, Nuventive Improve, to submit assessment reports (ROARs and Follow-Up reports).

Prior to 2025, Administrative Units created Goals to evaluate and report on each fiscal year. Beginning in 2025, administrative units are strongly encouraged to identify goals that are SMART or outcome-focused and that can be assessed for continuous improvement over a few years, rather than goals/outcomes that focus on task completion in one year. Administrative units' goals/outcomes and the associated plans for assessment are housed in in Nuventive Improve. Administrative goals/outcomes aim to assess services the units provide, whether that be efficiency of a service, satisfaction of end-users/stakeholders, etc., or learning-related outcomes that may result from trainings and other capacity building efforts of the unit. Ultimately, results obtained via this assessment process should be used to affect unit improvement.

Assessment Purpose

The assessment process focuses on the outcomes, rather than the outputs of work processes, thus the process is intended to measure outcomes (results achieved) rather than outputs (tangible work products or activities completed to help achieve outcomes). The following examples from the University of Wisconsin-Madison help illustrate the difference between output and outcome:

Outputs – Activities	Outcomes
The program trains and empowers community volunteers.	Community volunteers have knowledge and skill to work effectively with at-risk youth.
Program staff teach financial management skills to low-income families.	Low-income families are better able to manage their resources.
The camp experience provides leadership development opportunities for 4-H youth.	Campers, aged 12-15 years of age, learn new leadership and communication skills while at camp.
An annual conference disseminates the latest forage research.	Forage producers in Pasture County know current research information and use it to make informed decisions.

Outcome data is generally superior in providing information that can be used to improve programs and services. Assessment serves institutional effectiveness through its four main purposes:

- Improves programs/services through assessment results that indicate areas for change.
- Informs students, faculty, staff, and other stakeholders of the state of a program/service and its impact.
- Validates that a program/service is accomplishing what it says it is accomplishing through a demonstration of assessment results.
- Supports campus-decision making processes, strategic planning, program review and additional accountability activities such as SACSCOC reaffirmation and re-accreditation of academic programs by professional accrediting bodies (adapted from Selim et al., 2005b and Virginia Commonwealth, 2002).

The practice of outcomes assessment at the course and program level at Rice aligns with the American Association for Higher Education (AAHE) Nine Principles of Good Practice in Assessing Student Learning (Bauer, 2003):

- Principle 1: Assessment of student learning begins with educational values.
- Principle 2: Assessment is most effective when it reflects an understanding of learning as multidimensional, integrated, and revealed in performance over time.
- Principle 3: Assessment works best when the programs it seeks to improve have clear, explicitly stated purposes.
- Principle 4: Assessment requires attention to outcomes but also, and equally, to the experiences that lead to those outcomes.
- Principle 5: Assessment works best when it is ongoing, not episodic.
- Principle 6: Assessment fosters wider improvement when representatives across the educational community are involved.

- Principle 7: Assessment makes a difference when it begins with issues of use and illuminates questions that people really care about.
- Principle 8: Assessment is most likely to lead to improvement when it is part of a larger set of conditions that promote change.
- Principle 9: Through assessment, educators meet responsibilities to students and the public.

Understanding the Assessment Process

The ROAR process is a multi-phase, multi-element process, which can be described in three phases: 1--Planning, 2--Implementation, Assessment, Analysis, Reporting and finally 3--Improvement (i.e., Follow Up). The process is characterized in Figure 1 below (Rice University Annual Educational Assessment Process):

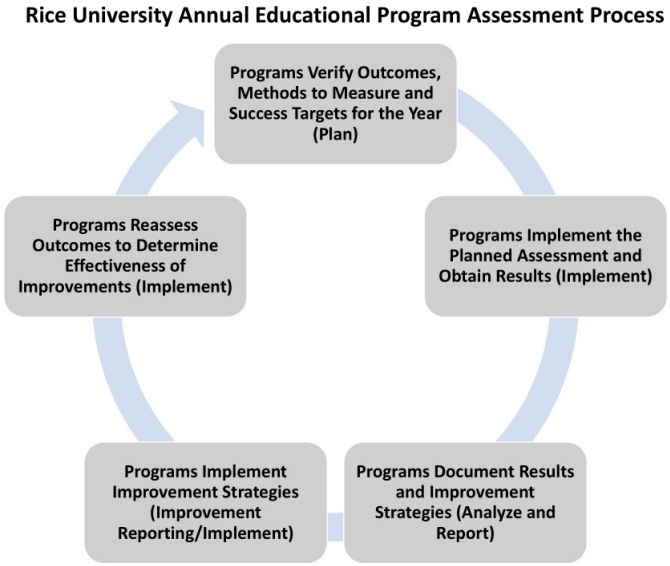

Figure 1: Rice University Annual Educational Assessment Process

Phase 1: Planning (Step 1 in Figure 1)
In Phase 1, assessment plans are designed/updated by programs (or goals set by administrative units) and reviewed by IE. As of Spring 2026, these plans are submitted to IE and then entered by IE staff into the assessment platform Nuventive Improve. For those who find a physical template for planning purposes useful, separate planning templates for academic programs and administrative units are provided. See the Appendix for a blank assessment plan template that includes definitions of the major elements.

Phase 2: Implementation, Assessment, Analysis, and Reporting (Step 2 and 3 in Figure 1)

In Phase 2, the planned assessment activities are implemented, assessment occurs and is followed by an analysis of the results. Historically, results of assessment have been reported on an individual ROAR template and is often referred to as an "initial ROAR." As of Spring 2026, IE leverages an online platform (Nuventive Improve) to store and manage ROAR assessment material, including the submission of initial ROARS. For those who would like to use the template version of the Initial ROAR reports for planning purposes, see Appendix to review a blank ROAR template. The initial ROAR report should include a discussion of proposed actions for improvement if the assessment targets were not achieved. Even if assessment targets were achieved, a program may still choose to identify improvement actions (and is encouraged to do so).

Phase 3: Improvement (Follow Up Report; Step 4 and 5 in Figure 1)
In Phase 3, programs implement planned improvement actions and document those actions on the Follow Up report (see Appendix to review a blank Follow Up Report template). As of Spring 2026, Follow-Up Reports are also entered into the online assessment platform, Nuventive Improve. Proposed changes are implemented and follow-up results reported. This part of the process completes the planning and evaluation cycle by looking back over the change proposals of the previous cycle(s) and discussing improvement actions that have been taken. These actions may be large (e.g., making notable changes to the curriculum, or a process for administrative units) or more minor but still important (e.g., adding an engagement activity to correspond with a specific lesson or updating discussion prompts, or refining training process in administrative units). The report generated for this phase of the process serves as documentation that the University is "closing the loop" by using assessment results for improvement. It is important to remember that SACSCOC only considers discussions of improvement actions that have been taken as compliant with the expectation that assessment results are used for improvement. If a planned improvement was not implemented, a Follow Up Report should still be submitted that includes: an explanation of why the planned improvement action did not occur, when the planned improvement will be implemented, and any other assessment-data inspired improvement actions that were implemented instead.

A complete assessment cycle for each program includes a documented assessment plan, an initial ROAR for each PLO (which is reassessed every two years on the new schedule) and a Follow Up Report documenting improvement actions taken based on systematically gathered and analyzed assessment results. SACSCOC expects institutions to demonstrate a completed assessment cycle for each program every three years. IE staff review all submitted ROAR documents and provide feedback and assistance to programs. Individual faculty leading assessment efforts for programs are given access to their program ROAR report archive in Nuventive and the archive of ROAR materials pre-dating 2020 which are stored in Box. IE also provides regular reminders of upcoming ROAR deadlines, offers 1:1 and small group custom educational sessions on outcomes assessment and ROAR reporting, and general educational sessions for any faculty and staff interested in learning more about outcomes assessment.

<u>How to proceed If you are leading ROAR activities for your program/unit</u>

Planning is the first phase of the ROAR process, and its importance is not to be underestimated as it directs a program's assessment activities. The following are recommended early steps in preparing for assessment:

- As of Spring 2026, familiarize yourself with the updated cycle of assessment: Even numbered PLOs/Goals will be assessed and reported on during even numbered years, odd numbered PLOs/Goals will be assessed/reported on during odd numbered years. Follow-Ups reports will be completed every year – at least one based on systematically gathered and analyzed assessment data. This schedule ensures that program assessment meets the accreditor requirement of a complete cycle of assessment for each PLO within three years.
- Review current assessment plan (if one exists) and published PLOs in the GA (administrative units should review the current unit ROAR Goals/outcomes)
- Review past ROAR reporting materials (ask IE for access if you don't have access)
- Review PLOs/Goals that are already housed within Nuventive Improve, as well as plan content (see Rice IDEAS website, ideas.rice.edu, for further information)
- Any programs who want to update PLO wording should first update PLOs with the Office of the Registrar (OTR) – changes are then relayed by OTR to IE.
- Units should contact assessment@rice.edu if plan content and/or Goals need updating

- Inventory existing and needed direct assessment methods (including rubrics, exam items; administrative units will want to review direct and indirect methods which may include surveys or tracking data)
- Review/update the desired target for assessment achievement
- Programs should talk with faculty who will be assessing the various PLOs so they are aware of expectations (administrative units should ensure that those responsible for implementing strategies to achieve goals, and collecting data to measure goal attainment are aware of the responsibility, timeline, and have needed resources)

For Programs

Sometimes updates to PLOs or the Assessment Plan are needed. A crucial part of the assessment plan is the intended outcome statements, or PLOs. The PLOs are statements of demonstrable knowledge, skill, or ability (KSAs) that students will possess upon completion of a program. The PLOs used on assessment plans should be the same ones that are published in the GA. PLOs should be reviewed on a regular basis and updated as is needed. To update PLOs, you may consider asking the following questions:

- What KSAs should students demonstrate as they complete the program?
- What evidence can be used to measure student KSAs?
- How can the quality of the KSAs be confirmed?
- Will data be obtained that can be used to improve the program?

Remember that a ROAR report focuses on assessment of a PLO conducted in a given academic year based on assessment activities implemented during the same academic year. Be sure that when updating PLOs, to review methodology to ensure that an appropriate method is in place within the curriculum (or can be developed) to directly measure student produced work as a measure of outcome attainment. PLOs should focus on measuring cognitive skill with active verbs at levels

appropriate for the specific degree program, rather than actions or outputs. The use of Bloom's taxonomy can help further guide your efforts to update PLOs.

For administrative units

Your unit will set goals/outcomes annually for the Fiscal Year. The Goals should be specific and measurable using direct or indirect methods. Goals should be aligned with the mission or operational objectives of your unit. Although it may be tempting to include goals related to accomplishing specific tasks, avoid task completion oriented goals (i.e., goals for which you can check a box indicating that the task was/was not completed) and instead, focus on goals that may be measured for improvement (such as an increase in end user satisfaction, improved time to completion for work orders, and the like). For goals associated with larger projects, consider setting incremental goals for each year that represent project phases for which success achieving that incremental goal may be measured. Goal statements should thus be focused on outcomes (a.k.a. Intended Outcome Statements) or what you would like to achieve. Use the SMART goals model to design your goals – a goal should be specific, measurable, achievable, realistic and time-bound. More information about SMART goals is in the Appendix.

Writing Intended Outcome Statements

Intended outcomes statements have a prescribed format and are stated in terms of what students or other stakeholders will accomplish. When writing intended outcomes statements focus on the following:

- Focus on a result to be achieved rather than an action (program, process, etc.) to be implemented.
- Write PLOs/outcome statements using action verbs (see Bloom's Taxonomy discussion below).
- Programs should write PLOs from the learner perspective (administrative units should focus on specific, measurable operational goals/objectives in writing outcome statements).
- Write PLOs/outcome statements in specific, measurable terms.
- Strive to focus on a single achievement/accomplishment in the PLO/outcome statement rather than multiple (to more clearly align measures)

The ABCD Outcome Writing Model is a simple structure to help write student and program learning outcome statements.

<u>The ABCD Outcome Writing Model</u>

You may wish to consider using the ABCD Outcome Writing Model (Henrich et al., 1996), especially if you are new to outcome writing. ABCD is a commonly used outcome writing model across a variety of educational institutions. ABCD is an acronym whereby "A" represents the audience—the learners of focus, "B" represents the observable and measurable behavior (or knowledge, skill or ability) the learner is expected to perform/exhibit, "C" represents the condition, circumstances or

context in which the learning occurs, and "D" represents degree of mastery (how well must the behavior be performed). According to the model, a well-written outcome includes all four parts (i.e., ABCD). The structure below may be used to help you write measurable outcomes (Caretta and Rice, 2010):

Students/stakeholders [audience] (A) will comprehend/analyze/demonstrate/create [what] (B) following/during/under these circumstances/conditions [where] (C) [at this level of achievement; degree] (D)

Example: Economics Mathematical Economic Analysis (MTEC) BA students will demonstrate the use of mathematical skills, including the elements of multiple variable calculus, linear algebra and optimization techniques, and other mathematical methods, in technical economic analyses at an advanced level.

Harvard Medical School reminds writers to use active verbs and aim for higher order thinking/demonstrations of higher order cognitive ability. Bloom's Taxonomy offers a structure to help select active verbs to use in formulating learning outcomes and to evaluate if the selected active verb is appropriately higher order.

Bloom's Taxonomy

Historically, discussions about student learning have been guided by a taxonomy of learning that has come to be known as Bloom's taxonomy (Bloom, 1956). This taxonomy is a hierarchical structure representing six levels of thinking and learning skills that range from basic learning objectives such as knowledge of content through higher- order learning such as synthesis, evaluation, and creativity. Bloom's taxonomy formed the basis for early work on the development of instructional objectives for classes and curricula (Excerpted directly from University of West Florida, n. d., para. 1). Instructional objectives were originally described as things (knowledge, understanding, content, facts) that could be delivered during a lecture or presented in written text. In contrast, student learning outcomes are described using concrete verbs (behaviors that can be observed in the student) rather than nouns. A more contemporary understanding of Bloom's Taxonomy for outcome assessment purposes is presented by Anderson and Krathwohl (2001) who adapted Bloom's model to reflect the needs of today's outcome-oriented language by changing nouns to active verbs. Most notably, knowledge has been converted to remember. In addition, the highest level of development is created rather than evaluate. (Preceding paragraphs excerpted and modified from University of West Florida, n. d. para. 1; Smythe and Halonen, 2009). Please see the Appendix for a list of active verbs using Bloom's Taxonomy.

Determining Assessment Methodology

Assessment methods outline what data will be collected, from what sources, using what methods, by whom, and in what approximate timeframe. The criteria below should be thoroughly considered when choosing an assessment strategy (adapted from Hoy, 2006; Palomba and Banta, 1999):

- Consider the relationship of the proposed method to the assessment question. Does the

method have the ability to answer the assessment question and provide the needed data? Some assessment methods are better than others at answering the same question. An example of using a measure that does not consider the assessment question would be to use a student satisfaction survey to measure student knowledge. Measures of student knowledge (i.e., PLOs/SLOs) should use direct measures only.

- Consider the reliability of the assessment method. Reliable measures are ones that produce consistent results through time. Measurement errors can occur through the individuals responding, through the administration and scoring of the instrument and through the instrument itself. Some errors can be minimized or avoided by ensuring that the instrument is well written and without ambiguity.

- Consider the validity of the measurement instrument. Validity refers to the instrument's ability to measure what we want it to measure. Are we measuring what we say we are measuring?

- Consider the timeliness and cost effectiveness of the method. When choosing an assessment strategy, consider if the measure will allow your unit to conduct assessment within the confines of the academic year. Also consider if your department has the funds/personnel on hand to conduct the assessment using the method in question, and realistic questions like, is the class we plan to assess the outcome in, going to have enough students to run in the planned semester?

- Consider the significance of the data that will be produced. Will the data that will be produced be good enough to use for decisions and improvement? Is the data going to be useful for making improvements at the unit level (as opposed to improvements that must be made on a larger scale)? It is best to draw conclusions about outcome attainment based on the entire sample of possible students (e.g., all instances of the student work prepared and evaluated for the PLO under consideration, such as a paper) however if you choose to evaluate a sample, ensure that the sample reviewed is representative of the whole in all ways, and also represents a strong majority (preferably close to 75% of all available student work).

- Consider useability of the results. Think about whether the assessment strategy will provide results that are clear to understand, interpret, and then use for continuous improvement purposes. Determine if the results can provide the type of information your unit needs to promote change.

Also consider the methodological "musts" listed below. The selected methodology should meet each of these essential points:

- Assessment methodology must evaluate the extent to which the PLO or administrative goal is achieved, not if steps were taken to implement actions related to the PLO.

- Assessment methodology intended to measure PLOs/student learning must examine a

student work product (a performance, a written paper, exam questions, etc.). This is referred to as using a direct measure of student learning. Examples of direct measures include:

- Exams
- Portfolios
- Research papers, thesis & dissertations
- Exhibitions and demonstrations
- Oral exams or presentations

Generally, a rubric or set of established criteria with a rating scale is used to directly evaluate the student created/presented works above, in order to determine if the PLO was achieved and to what extent. Note that administrative goals may be measured using direct methods or indirect methods (indirect methods are those which are based on data other than actual work produced by learners and which may suggest outcome attainment but are not definitive, such as survey data that is opinion or perception based, tracking data such as the number of users or hours completed, course evaluations, graduation rates, and the like).

When selecting embedded assignments to use in outcomes assessment activities, consideration should be given to the in-course evaluation method of student work used for assessment. It may be necessary to modify the way in which the assignments submitted for course requirements are evaluated (i.e., the grading method) or consider adding an additional method so the embedded course work may be suitable for assessment purposes. The use of course assignments and the matter of "grades" is discussed more in the next section. You may also consider employing more than one method of assessment for each intended outcome. Finding that two assessment methods produce similarly positive (or negative) results lends some degree of validity for the results and conclusions drawn from those results. Using multiple measures to increase the degree of validity is referred to as triangulation of data.

Avoid Using "Grades" and Scoring with Undefined Criteria

Please note that course grades may not be used to measure PLO attainment.

Assessment is not a substitute for grading and grading is not a substitute for assessment. The primary focus of assessment is to use embedded student work to obtain a snapshot of student performance and achievement related to the PLO, using a specific data set. Ideally, assessment of the various PLOs can be done over time and all results used for program continuous improvement purposes. Course grades should not be used because they provide an overall evaluation of a student's performance in a course rather than learning attainment for a specific program learning outcome. Outcome assessment is focused on evaluating a single outcome at a time; therefore, course grades should not be used as the criteria for success in outcomes assessment.

Similarly, grading systems that employ unstated or undefined criteria, such as "it feels like a B" (Walvoord and Anderson, 1998) should not be used as the sole criteria for success for the same reason as overall course grades should not be used. These grading methods may be used as a criterion only in conjunction with more detailed secondary criteria. Similarly, pass/fail designations are also not preferred—if this approach is used for evaluation of student work, consider

documenting what characteristics the student work must have to earn a "pass" or "fail" designation and associating each with a numeric scale. In this way, the evaluation can become more reliably used over time as different raters can utilize the characteristics in the determination of learning outcome attainment, which will be more useful for continuous improvement purposes.

Set a Target for Achievement

In addition to establishing the outcome and describing the method to be used for assessment, it is important to clearly define the desired level or target of achievement by the population to be measured. Generally, a target for achievement will establish what amount (e.g., a percentage) of the whole group assessed should demonstrate a certain level of performance. For a PLO, the target for achievement could be something like 80% of assessed student papers will earn a rubric rating score average of 3 (using a rating scale of 1-4 where 4 demonstrates highest achievement). For an administrative goal, the target may be similar to the PLO target if there is a learning outcome being assessed. For operational outcomes, the target would focus on the level of desired attainment as related to the specific operational achievement. So, if the outcome was to improve overall end user satisfaction with a provided service, and it was being measured with an item on a survey that asked end users to rate their overall experience with the provided service, the target for achievement could identify the percentage of respondents to the survey who indicated that they were satisfied or very satisfied (using a satisfaction scale of very dissatisfied to very satisfied). An administrative goal focused on reducing response time to service tickets that is measured by timestamps in an electronic ticketing system, could compare response time averages from a prior year, with those obtained for a current year, and set a target to reduce response time by eight hours. The target selected should be appropriate for the outcome/goal and the method and be appropriate to demonstrate that improvement efforts are being made.

Next: Implement Your Assessment Activities! Be sure to follow the assessment plan—if anything needs to change from what was planned, then be sure to document the change somewhere to help when you write your report (the ROAR).

Reporting Assessment Activities and Results (a.k.a. the ROAR!)

As of Spring 2026, programs and units will enter ROARs into the online assessment platform, Nuventive Improve. When documenting the results of assessment activities, begin with a description of the method actually used (if it differs from the planned method) and detail the data collected. In the detail of the data collected, be sure to include:

- a description of the group that was assessed and total number of students whose work was assessed – be sure to also specify the course(s) in which assessment occurred
- a description of the data collected (i.e., the total number of student works assessed, or surveys collected/process assessed for administrative units, and detailed rationale for any sampling and the detailed sampling method)
- aggregated discussion of the results obtained, and
- a description of how the program/unit examined the results.

The results report should also include brief discussions of:

- analysis of the extent to which the intended outcome was achieved (i.e. achieved, not achieved, or partially achieved)—be sure to include the comparison of results obtained to the planned/desired target for outcome attainment as shown in the plan.
- mention who collected, analyzed and interpreted the assessment results (e.g. assessment committee, department chair, outside consultant, entire staff or faculty of a unit), and
- provide an interpretation of the results (e.g., explain what the data means, discuss any anomalies in the results and try to account for them and also note any extenuating circumstances affecting the results)

It is expected that results of assessment and subsequent interpretation will be disseminated to members of the unit and that feedback will be solicited from the unit team members including those providing unit oversight, especially in cases where improvement is warranted. Once feedback on the results is obtained from unit members, improvement actions and strategies can be proposed. Be sure to share planned improvement actions with team members and decision-making persons and engage them in a discussion about a timeframe for implementation and resources needed to implement the change(s).

After you complete your results report, be sure to select the appropriate drop-down menu item to indicate the extent to which results were achieved (fully, partially, not at all, inconclusive).

Improvement Plan—Required

In cases where outcomes are not achieved or partially achieved, be sure to include a brief discussion of planned improvement actions. The improvement should directly relate to the assessment results. Even if you meet the target and if the desired outcome is achieved you are strongly encouraged to still include an improvement plan since this helps guide continuous improvement activities.

Possible Improvement Plan actions include:

- Changes to curriculum (e.g. additional readings, assignment changes, updating discussion prompts)
- Changes to measurements or assignments
- Adding additional student learning supports (e.g. tutoring, supplemental instruction, peer models)
- Adjusting instructional methods
- Enhancing unit operations (Administrative)
- Changes to funding/resource allocation (Administrative)
- Improve staff development (Administrative)
- Improve outreach (Administrative)

It is not acceptable to simplify the target for achievement to increase likelihood of outcome attainment. After you complete your Improvement Plan discussion, please select the appropriate drop-down menu item to indicate improvement actions to be taken.

Follow Up Report

As of Spring 2026, programs and units will enter Follow Up Reports into the online assessment platform, Nuventive Improve. The final step in the assessment cycle is to implement changes based on assessment results, in order to improve programs and services. At Rice, the Follow Up Report is used to document the actions taken/changes made based on the use of assessment results. These reported improvements could be based on any systematically gathered and analyzed assessment data, including institutional data, previous ROARs, or other relevant programmatic/unit data. Be sure to provide a detailed discussion of the actions you took, based on the assessment results you obtained, and if you re-assessed after implementation of improvements, include those follow-up assessment results. Remember also that if you decide to make changes to how you assess (e.g., improvements to method), to update your assessment plan. Realizing that changes are needed to how assessment is conducted is a worthwhile conclusion and may be documented on the Follow Up report.

The Assessment Cycle

Outcomes assessment is a cycle. Be sure to review the updated (as of Spring 2026) assessment cycle (even numbered PLOs/goals during even numbered years, odd during odd years, follow-ups every year), and complete assessment activities and required reporting to IE on schedule. Remember that academic program PLO ROARs and Follow Up reports are due June 1 annually (and are based on assessment activities and data collected in the prior academic year, so, the report is submitted at the end of the academic year in which assessment occurred). Administrative assessment follows the fiscal year cycle, and administrative unit ROARs and Goals for the upcoming fiscal year are due annually July 1. According to SACSCOC a completed assessment cycle, to include re-assessment and report of each PLO is expected within a three-year timeline. Fortunately, the new odd/even scheduling approach, implemented AY26, helps to ensure that programs will meet this expectation, with on-time reporting. SACSCOC also requires that institutions demonstrate how programs and units are using systematically gathered and analyzed assessment data for improvement purposes. The Follow Up report, submitted on time, helps ensure that this requirement may be demonstrated.

Section IV: Obtaining Support for Assessment Work

Consider reviewing the resources posted in the Outcomes Assessment course posted by IE and available in Canvas. Nuventive Improve specific resources are also available within the platform. As of spring 2026, the Canvas Outcomes Assessment training course is available for all assessment personnel focusing on Nuventive, and in summer 2026, will include assessment fundamentals asynchronous educational materials. Please contact assessment@rice.edu for further assistance with the Canvas course.

IE staff are happy to answer ad hoc questions from you or your team, to join you in a meeting to provide education or answer questions, and to prepare and deliver planned professional

development sessions for your team. IE can facilitate work sessions in a lab format for your team working with updating or preparing assessment materials. IE offers regularly scheduled educational sessions in short format which are open to all members of the Rice community. Please let us know how we can help!

APPENDIX

Bloom's Taxonomy Action Verb List
(from Dartmouth Center for the Advancement of Learning. Retrieved 05/22/24 at chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://dcal.dartmouth.edu/sites/dcal.prod/files/center_for_advancement_learning/wysiwyg/blooms_taxonomy_action_verbs_1.pdf)

Bloom's Taxonomy Action Verbs

Definitions	Knowledge	Comprehension	Application	Analysis	Synthesis	Evaluation
Bloom's Definition	Remember previously learned information.	Demonstrate an understanding of the facts.	Apply knowledge to actual situations.	Break down objects or ideas into simpler parts and find evidence to support generalizations.	Compile component ideas into a new whole or propose alternative solutions.	Make and defend judgments based on internal evidence or external criteria.
Verbs	• Arrange • Define • Describe • Duplicate • Identify • Label • List • Match • Memorize • Name • Order • Outline • Recognize • Relate • Recall • Repeat • Reproduce • Select • State	• Classify • Convert • Defend • Describe • Discuss • Distinguish • Estimate • Explain • Express • Extend • Generalized • Give example(s) • Identify • Indicate • Infer • Locate • Paraphrase • Predict • Recognize • Rewrite • Review • Select • Summarize • Translate	• Apply • Change • Choose • Compute • Demonstrate • Discover • Dramatize • Employ • Illustrate • Interpret • Manipulate • Modify • Operate • Practice • Predict • Prepare • Produce • Relate • Schedule • Show • Sketch • Solve • Use • Write	• Analyze • Appraise • Breakdown • Calculate • Categorize • Compare • Contrast • Criticize • Diagram • Differentiate • Discriminate • Distinguish • Examine • Experiment • Identify • Illustrate • Infer • Model • Outline • Point out • Question • Relate • Select • Separate • Subdivide • Test	• Arrange • Assemble • Categorize • Collect • Combine • Comply • Compose • Construct • Create • Design • Develop • Devise • Explain • Formulate • Generate • Plan • Prepare • Rearrange • Reconstruct • Relate • Reorganize • Revise • Rewrite • Set up • Summarize • Synthesize • Tell • Write	• Appraise • Argue • Assess • Attach • Choose • Compare • Conclude • Contrast • Defend • Describe • Discriminate • Estimate • Evaluate • Explain • Judge • Justify • Interpret • Relate • Predict • Rate • Select • Summarize • Support • Value

Institutional Effectiveness

Rice Outcome Assessment Report (ROAR) - Academic Program Results Report Planning Template

The following template and its sections are intended to mirror the ROAR report elements set-up shown in the Nuventive Improve platform.

Use the following template to plan your ROAR entry, then copy and paste the contents into the corresponding sections in Nuventive. Complete one planning template per PLO for which you will report results. Once the initial ROARs have been entered and saved in Nuventive, please notify the IE team. *For help with the Nuventive Improve platform, contact assessment@rice.edu.*

Department/Academic Program:

Primary Contact (Name/Phone/E-mail):

Academic Year:	
Program Learning Outcome *Specific program learning outcome(s) listed in the General Announcements*	
Methods *What kind of direct assessment did you use? (i.e. describe how you selected the sample of work that was assessed, the rubric used, etc. Best practices and SACSCOC requires the use of direct assessment.)* *If you also used indirect assessment, please describe that as well.*	
Results and Conclusions *Describe with specificity the population assessed (disaggregating as needed – data should provide aggregate results for students in the specific program being assessed*	

rather than all students enrolled in a course; similarly a program/course offered in multiple modalities must be disaggregated by modality), and the results received. Please include an analysis of these results and how they are being interpreted. <u>An obviated outcome achievement statement should be included that directly correlates to your set standard.</u>	
How will the results be used for improvement purposes?	*In Nuventive Improve, this section will include a section for you to provide a short narrative explanation, and a drop-down with the options shown below – include your narrative and make a drop down selection in Nuventive:* Changes to Curriculum Changes to Instruction Enhance Faculty Development/preparation Add Student support such as mentoring Add student learning resources Fully Achieved- Plan to reassess Improve outcome design Improve methodology design Improve outcomes assessment processes Enhance program operations that impact learning
Expand on how you will use the results for improvement (Results, Timeline, Re-assessment) *Based on your results, what changes (if any) will you make?* *What is your timeline to make these changes?* *What is your timeline for assessing the impact of these changes?* *For Fully Achieved outcomes, it is strongly*	

recommended to identify how the results will be used for improvement. If reassessing for consistency, please state so.	

Institutional Effectiveness

Rice Outcome Assessment Report (ROAR) – Administrative Unit Results Report Planning Template

The following template and its sections are intended to mirror the ROAR set-up shown in the Nuventive Improve platform.

Use the following template to plan your ROAR entry, then copy and paste the contents into the corresponding sections in Nuventive. Complete one planning template per goal/outcome for which you will report results. Once the initial ROARs have been entered and saved in Nuventive, please notify the IE team.
For help with the Nuventive Improve platform, contact assessment@rice.edu.

Department/Unit:

Primary Contact (Name/Phone/E-mail):

Fiscal Year:	
Goal/outcome *Specific Goal/Outcome as stated in Nuventive Improve and in assessment plan*	
Methods- How did you evaluate your achievement of the goal/outcome (whether or not you accomplished this goal/outcome)? What kind of information did you collect? Please describe. *Describe how you measured goal/outcome achievement (what assessment was used, where/when it occurred, who/what was assessed, rubric or assessment tool explanation, etc.). Please note any changes made to methodology as compared to your original assessment plan and detail the reason for those changes (e.g. assessing at a different time of year,*	

rubric/assessment tool change, etc.)	
Results and Conclusions- What were the results of your assessment? *Describe with specificity the population/process assessed, and the results received. Please include an analysis of these results and how they are being interpreted. <u>An obviated outcome achievement statement should be included that directly correlates to your set standard.</u>*	
How will the results be used for improvement purposes?	*In Nuventive Improve, this section will include a section for you to provide a short narrative explanation, and a drop-down with the options shown below – include your narrative and make a drop down selection in Nuventive:* Enhance unit operations that impact services provided Changes to funding and resource allocation Improve outreach to community/community engagement Improve staff development Fully Achieved–Plan to reassess Improve outcome design Improve methodology design Improve outcomes assessment process
Expand on how you will use the results for improvement (Results, Timeline, Re-assessment) *Based on your results, what changes (if any) will you make? Expand on the selected improvement from the options in the section above.* *What is your timeline to make these changes?* *What is your timeline for assessing the impact of these changes?*	

For Fully Achieved outcomes, it is strongly recommended to identify how the results will be used for improvement. If reassessing for consistency, please state so.	

Rice University ROAR Follow-Up Report Template- Academic AND Administrative

Program or Unit:_____

The following template and its sections are intended to mirror the Follow-Up Report set-up shown in the Nuventive Improve platform. **Use the following template to plan your Follow-Up Report entry, then copy and paste the contents into the corresponding sections in Nuventive. Once the Follow-Up Report has been entered and saved in Nuventive, please notify the IE team. For help with the Nuventive Improve platform, contact assessment@rice.edu**

ACADEMIC		ADMINISTRATIVE	
Improvement Name *(Name the improvement using short descriptive language)*		**Improvement Name** *(Name the improvement using short descriptive language)*	
Describe one improvement your program made during the past academic year that was based on systematically gathered and analyzed assessment data. This can be data of any kind, including institutional data, previous ROARs, or other relevant programmatic data *(Improvements documented on this report can be tied to any previous systematically gathered and analyzed assessment data. It is important to provide contextual information (including data analyzed) about the improvement as well as details of the improvement; e.g., previous ROAR data showed a need for revision of instruction for X assignment). Reassessment data (if available) can be reported here to show effects of the implemented improvement.*		**Describe one improvement your unit made during the past fiscal year that was based on systematically gathered and analyzed assessment data. This can be data of any kind, including institutional data, previous ROARs, or other relevant unit data** *(Improvements documented on this report can be tied to any previous systematically gathered and analyzed assessment data. It is important to provide contextual information (including data analyzed) about the improvement as well as details of the improvement; e.g., previous ROAR data showed a need for improvement of X process). Reassessment data (if available) can be reported here to show effects of the implemented improvement.*	
Academic Year Implemented *(Year the improvement was implemented--improvement(s) should have been made during the reporting year)*		**Fiscal Year Implemented** *(Year the improvement was implemented--improvement(s) should have been made during the reporting year)*	
Systematically gathered data that showed a need for improvement. *(Briefly describe the assessment findings that prompted the improvement. Improvements could have been made based on any systematically gathered and analyzed data from the program. If the need was gleaned through a previous ROAR assessment, include contextual information (assessment year) and details of the previous ROAR results.)*		**Systematically gathered data that showed a need for improvement.** *(Briefly describe the assessment findings that prompted the improvement. Improvements could have been made based on any systematically gathered and analyzed data from the unit. If the need was gleaned through a previous ROAR assessment, include contextual information (assessment year) and details of the previous ROAR results.)*	

Rice University ROAR Academic Program Assessment Plan Template

Program:_____, AY__

****PLEASE COMPLETE AND RETURN TO INSTITUTIONAL EFFECTIVENESS. IE WILL ENTER INTO NUVENTIVE IMPROVE****

Learning Outcomes *(Below each numbered PLO, insert title of PLO to match GA PLO wording; add columns after PLO 4 for each additional PLO)*	**PLO 1** **PLO title:**	**PLO 2** **PLO title:**	**PLO 3** **PLO title:**	**PLO 4** **PLO title:**
Embedded location *(Where? Describe where the students' work to be assessed is found (e.g., Capstone Course)*				
Primary type of student work assessed (Materials) *(What? List actual specimen(s) of student work being used for the direct assessment–paper, presentations, exams, etc)*				
Measure *(How? Describe how the student work will be directly assessed, e.g., with a rubric or other set of criteria, by a faculty member or faculty panel of raters, etc.)*				
Standard/Criteria for Success *(indicate the target level of desired performance by students on the directly assessed work to determine outcome attainment; e.g., at least 80% of students' work will earn an average rubric rating of 3 out of 4 on the specified rubric dimension)*				
Who conducts the assessment and analyzes the results? *(using job titles, describe who conducts the assessment and evaluates results; e.g., instructor of record, a faculty rater panel, an external expert panel of industry partners.)*				
Timeline *(When will the assessment of this PLO occur; e.g., Initial assessment for PLO 3 occuring in Spring 2027, as odd numbered PLOs are reported in odd numbered years.)*				

Rice University ROAR Administrative Unit Assessment Plan Template

Office:_____, FY__

****PLEASE COMPLETE AND RETURN TO INSTITUTIONAL EFFECTIVENESS. IE WILL ENTER INTO NUVENTIVE IMPROVE****

Goals/Outcomes *(Below each numbered Goal, insert Goal Statement; A minimum of 3 Goals is reccommended ; add columns after Goal 4 for each additional Goal)*	**GOAL 1**	**GOAL 2**	**GOAL 3**	**GOAL 4**
Goal Statement *(Identify a specific goal (an outcome your unit wants to achieve) in the FY. These should be concise, specific, and measurable. Avoid task completion goals—focus on services or processes that can be improved or assessed for quality (such as end user satisfaction, time to completion of a service, etc.). Goals should be aligned with the mission or operational objectives of your unit.)*				
Fiscal Year of Assessment *(When will the assessment of this goal will occur; e.g., Initial assessment for Goal 3 occuring in Spring 2027, as odd numbered Goals are reported in odd numbered years, and even numbered goals are reported in even numbered years.)*				
Action Steps *(Describe the steps your unit will take to work toward achieving this specific goal/outcome. Examples may include reviewing current procedures, conducting training, implementing new procedures, etc.)*				
Baseline *(Describe any baseline data you have concerning this goal/outcome. This data is important for any goals/outcomes assessing an increase/decrease in a measure, improvement of a process, etc. If no baseline exists, or if you are trying to establish a baseline, please state that.)*				
Measure/Method *(Describe how you will evaluate to determine if you meet your goal. Outline what data will be collected, from what sources, using what methods (survey, rubric, platform, etc), by whom, and when.)*				
Standard for success/Target *(indicate the target level of achievement to indicate Goal attainment; e.g., at least 80% of survey respondents will indicate high satisfaction with X service; Completion of X service will take 24 hours less than the previous FY, etc.)*				
Impact *(Explain why this goal is important for your unit/the impact this will have on your unit's work.)*				
Who conducts the assessment and analyzes the results? *(using job titles, describe who conducts the assessment and evaluates results; e.g., Director)*				

Glossary

This glossary contains terms and names that you may encounter in your assessment work. All definitions from Sweeny, 1994 unless otherwise cited.

Administrative Outcomes

Operational and specific statements derived from a unit's core functions that describe the desired quality of key services within an administrative unit and define exactly what the services should promote (modified from Selim et al., 2005, p. 19).

Aggregated Scores

The combined scores for a population of students often expressed as an average. Aggregating scores requires that all the scores be based on the same or equivalent assessments administered in uniform ways.

Alignment

The process of assuring that learning outcomes, local curriculum and instruction and the system of assessment all support and match each other.

Analytic Scoring

The use of specific criteria or features to evaluate and assign points to each essential part of a product or performance. Analytic scoring is diagnostic, allowing planning for specific remediation. (See Holistic Scoring for the alternative approach.

Annotated Rubric

The notes from an assessment development group, often after a field test and initial scoring, which explain the meaning of criteria or distinctions between the criteria on a rubric. Annotation is an important tool to increase scoring reliability and to train others to score consistently.

Anchors

Actual samples of student work, which illustrate the essential characteristics of work typical for each scoring level on a scoring rubric. Anchors can also be captured on video or audio tapes of performances or may be video or photographic images of a larger product. The top anchor is often called an "exemplar" as it represents exemplary work.

Assessment

A systematic and ongoing process of gathering and interpreting information to discover if programs/services are meeting intended outcomes/objectives and then using the information to enhance the programs/services (adapted from Virginia Commonwealth University, 2002 & Marchese, 1987).

Authentic

A characteristic of assessments that have a high degree of similarity to tasks performed in the real world. The more authentic the assessment, the less inference required to predict student success after graduation.

Benchmark
A standard by which something can be measured or judged (Lexico Publishing Group, n. d.).

Comprehensive
All dimensions of a learning goal with regard to scope, content, specificity, skills, and types of thinking required are addressed.

Competency
See Student Competency

Course-Level Assessment
Assessment to determine the extent to which a specific course is achieving its learning goals. (For comparison, see Program Assessment and Institutional Assessment.)

Criterion-Referenced Test
A measurement of achievement of specific criteria stated as levels of mastery. The focus is performance of an individual as measured against a standard or criteria rather than against performance of others who take the same test. See Standardized, Norm-referenced Tests.

Curriculum Mapping
A matrix showing the coverage of each program learning outcome in each course. It may also indicate the level of emphasis of each outcome in each course (from Bridgewater State College, n. d. as Course Mapping).

Cut Score
The number of points needed which represents the criteria for successful completion of an assessment task, such as eight out of 10, or the percent that must be attained to be determined as successful in performing an assessment task (e.g., 80%). Cut score also refers to the critical point for dividing scores into two groups in reference to some criterion. It is possible to set multiple cut scores from differing criterion (e.g., meets, does not meet and exceeds).

Direct Assessment
Method of gauging student achievement of learning outcomes through evaluation of student work products (Bridgewater State College, n. d.). For comparison, see Indirect Assessment.

Disaggregated Group
Any group of students within a population from which a group score is computed as a group separate from the total assessed population.

Documentation
Written descriptions, reports or summaries of the steps taken and the rationale for those actions.

Exemplar
Actual sample of student work that illustrates the essential characteristics of work typical of exemplary student work at the top scoring level on a scoring rubric. Several exemplars are desirable to promote creativity so that students see multiple products/performances are possible.

Embedded Assessment Methods
Methods in which evidence of student learning outcomes for the program is obtained from assignments in particular courses in the curriculum (Bridgewater State College, n. d.).

Formative Assessment
The assessment of student achievement at different stages of a course or at different stages of a student's academic career. The focus of formative assessment is on the documentation of student development early and over time. It can also be used to engage students in a process of reflection on their education (modified from Bridgewater State College, n. d. & California Polytechnic State University, n. d.). For comparison, see Summative Assessment.

Forced-Choice Assessment
Testing where responses to an item, questions or prompts are limited to a provided set of options. Scoring does not require judgment on the part of the scorer because there is one right answer to the item. Multiple choice, true/false, cloze, and matching are examples of forced choice/short answer assessments.

Generalizable
An instance when the score on one assessment can accurately predict a student score on a different assessment covering the same knowledge or skill. Generalizability across time is promoted by ensuring that assessments focus on general level concepts or strategies, not on facts, topics, or skills, which are found only at one level or in one class.

Goals
The general aims or purposes of a program and its curriculum. Effective goals are broadly stated, meaningful, achievable and assessable. Goals provide a framework for determining the more specific educational objectives of a program and should be consistent with program and institutional mission.

Holistic Scoring
Scoring based upon an overall impression (as opposed to traditional test scoring, which totals specific errors and subtracts points based on them). In holistic scoring, the rater matches his or her overall impression to the point scale to see how the portfolio, product or performance should be scored. Raters usually are directed to pay attention to particular aspects of a performance in assigning the overall score.

Indicator
A statistic that reveals information about the performance of a program or a student. For a statistic to be an educational indicator there must be a standard against which it can be judged. Educational indicators must meet certain substantive and technical standards that define the kind

of information they should provide and the features they should measure. The primary educational indicator is student performance; other secondary indicators include attendance, graduation, mobility, and dropout rates.

Indirect Assessment
Assessment that deduces student achievement of learning outcomes through students' reported perception of their own learning. May also be the opinions or thoughts of others about student knowledge, skills, attitudes, learning experiences, and perceptions. Examples of indirect measures include student surveys about instruction; focus groups; alumni surveys; employer surveys (modified from Community College of Aurora, n. d., California Polytechnic State University, n. d., & Bridgewater State College, n. d.0). For comparison, see Direct Assessment.

Indirect Measure of Learning Outcome
Students or others report their perception of how well a given learning outcome has been achieved (California Polytechnic State University, n. d.).

Institutional Assessment
Assessment to determine the extent to which a college or university is achieving its mission. (For comparison, see Course- level Assessment and Program Assessment.)

Institutional Effectiveness
A continuous set of the processes of planning, assessment and review aimed at ongoing improvement (Southern Association of Colleges and Schools Commission on Colleges, 2004).

Item
An individual question or exercise in a test.

Methods of Assessment
Tests and procedures used to measure student performance in meeting the standards for a learning outcome. These assessments must relate to a learning outcome, identify a particular kind of evidence to be evaluated, define exercises that elicit that evidence and describe systematic scoring procedures.

Non-Discrimination
Evidence that differences of race or ethnicity, gender, or disability do not bias results of assessment instruments or procedures.

Objective
Precise statement that specifies the performance or behavior a student is to demonstrate relative to a knowledge or skill. Objectives typically relate to lessons or units, not "big ideas" such as described by an outcome.

Outcome
An end result; a consequence (Lexico Publishing Group, n. d.). See Administrative Outcome and/or Student Learning Outcome.

Overall Performance Level
A combination of the cut-scores or proficiency levels of the various assessments used to determine whether students do not meet, meet, or exceed the standard set for a whole learning outcome. Different assessments may be given greater weight when determining an overall performance level. See Weighting.

Performance Based Assessments
A methodology requiring reasoning about recurring issues, problems and concepts that apply in both academic and practical situations. Students actively engage in generating complex responses requiring integration of knowledge and strategies, not just use of isolated facts and skills.

Pilot
A large-scale administration of an assessment, usually with several classes of students if not all students in a program, designed to detect any flaws in the assessment before the assessment is considered "done" and is fully implemented. See Field Test for contrast.

Portfolio
A purposeful collection of artifacts that demonstrate a student's development or achievement. Using portfolios as an improvement assessment tool requires the ability to score both individual works and the whole portfolio against a standard for each modified form (Sweeny, 1994 & Bridgewater State College, n. d.).

Proficiency Level
The equivalent of a cut score (on a forced-choice assessment) but for a performance/complex assessment. The proficiency level for a performance assessment is set by determining the required performance criteria (such as the required level on a rubric) for a specific grade level. Such a proficiency level could be achievement of all the criteria required for a scoring level, or it could be a set number of points achieved by combining scores for each feature on the rubric.

Program Assessment
Assessment to determine the extent to which students in a departmental program can demonstrate the learning outcomes for the program. For comparison, see Course-level Assessment and Institutional Assessment.

Prompt
In a narrow sense, a prompt is a statement to which a student responds in an assessment, often a reading passage, picture, chart or other form of information. In the fullest sense, a prompt is the directions that ask the student to undertake a task. Prompts should include the context of the situation, the problem to be solved, the role the student takes, and the audience for the product or performance.

Rationale
Written statements providing the reasons for steps taken and choices made.

Raw Score
The number of items that are answered correctly out of the total possible.

Reliability
Consistency or stability of assessment results. Of particular importance for performance assessment is inter-rater reliability. It is the estimate of the consistency of the ratings assigned by two or more raters because they agree on the criteria used to evaluate the performance.

Representativeness
A factor of performance tasks and of scoring criteria ensuring that the task and criteria focus on the significant elements, concepts and strategies in the outcome(s) assessed.

Rubric
A set of criteria specifying the characteristics of a learning outcome and the levels of achievement in each characteristic (Bridgewater State College, n. d.).

Score
The result obtained by a student on an assessment, expressed as a number. Each score is recorded as a positive number, with a larger numerical value implying a better result.

Scoring Rubric
A set of related scoring scales used for judging student work and awarding points to reflect the evaluation of the work.

Scoring Scale
Assessment criteria formatted as levels of quality ranging from poorest to best, used to judge student work on a single feature such as "clarity of main idea." Scales may combine several traits within a feature. Scoring levels on the scale are assigned points, each level specifying the characteristics of the quality of content or skills needed to attain the points.

Self-Assessment
Students reflect about their own abilities and performance, related to specified content and skills and related to their effectiveness as learners, using specific performance criteria, assessment standards, and personal goal setting. The intent is to teach students to monitor their own learning continuously.

Self-Efficacy
Students' judgment of their own capabilities for a specific learning outcome (Bridgewater State College, n. d.).

Southern Association of Colleges and Schools Commission on Colleges
One of several US Department of Education recognized institutional accrediting bodies of higher education institutions which traditionally accredits institutions in the southern states (AL, FL, GA, KY, LA, MS, NC, SC, TN, TX and VA).

Standardized, Norm-Referenced Test
A form of assessment in which a student is compared to other students. Results have been normed against a specific population (usually nationally). Standardization (uniformity) is obtained by administering the test to a given population under controlled conditions and then calculating means, standard deviations, standardized scores, and percentiles. Equivalent scores are then produced for comparisons of an individual score to the norm group's performance.

Standard Score
A score that is expressed as a deviation from a population mean.

Student Competency
Statement of broad knowledge, skills, attitudes, or behaviors that program majors should be able to demonstrate upon completion of the degree program.

Student Learning Outcomes
Operational statements of demonstrable knowledge or skill that students will possess upon completion of a program or course. These statements may be the student program competencies or more specific statements derived from the student program competencies.

Summative Assessment
The assessment of student achievement at the endpoint of their education or at the end of a course. The focus of summative assessment is on the documentation of student achievement by the end of a course or program. It does not reveal the pathway of development to achieve that endpoint, but rather provides an evaluative summary (Bridgewater State College, n. d.). For comparison, see Formative Assessment.

Target for Achievement
The specific level of outcome or goal achievement desired by the individual/program conducting assessment.

Task
A goal-directed assessment activity or project, which prescribes that the student uses their background knowledge and skill in a somewhat long-term process to solve complex problems or answer a multi-faceted question.

Triangulation
Involves the collection of data via multiple methods in order to determine if the results show a consistent outcome (California Polytechnic State University, n. d.).

Validity
The degree to which an assessment measures (a) what is intended, as opposed to (b) what is not intended, or (c) what is unsystematic or unstable, thus producing accurate, meaningful, and useful measures of the skills and knowledge it was designed to assess. The primary issue is content validity, which is whether an assessment and instructional program align (match) (Bridgewater State College, n. d. & Sweeny, 1994).

Validation
The process of developing, field testing, refining, piloting and refining assessment items, tasks, scoring tools, directions, etc. to increase validity, reliability, fairness and instructional usefulness.

Weighting
A method to combine the results of two or more assessments used in calculating the percent who meet the standard for a learning outcome. If some assessments are deemed more important due to the amount of time for completion or the number of items included in the assessment, etc. the cut-scores on those assessments may be given greater consideration or weight in determining the overall performance level.

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